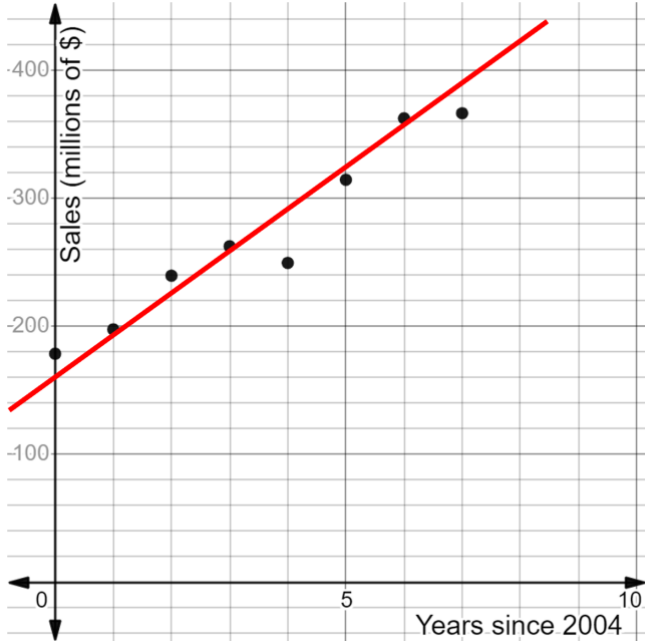


## Algebra 1 Unit 3 - Part 2 Assessment Guide

|                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p style="text-align: center;"><b>General Information</b></p>      | <p>The Unit 3 assessment is divided into two parts. Part 1 assesses benchmarks MA.912.AR.1.2 and MA.912.AR.2.2, and Part 2 assesses MA.912.DP.2.4 and MA.912.DP.2.6.</p> <p>The benchmarks MA.912.DP.2.4 and MA.912.DP.2.6 build on students' understanding of scatter plots from Grade 8 or Grade 7 Accelerated where students learned to construct scatter plots (MA.8.DP.1.1), describe patterns of association (MA.8.DP.1.2), and informally fit a straight line to a scatter plot with a linear association (MA.8.DP.1.3). Teachers familiar with the MAFS standards may not immediately notice that the benchmark MA.912.DP.2.6 states "given a scatter plot with a line of fit and residuals". Algebra 1 students are not required to plot the residual values. The draft item specifications include the assessment limit "Residual values must be represented graphically on the scatter plot or on a residual plot." Question 3 adheres to this limit.</p> <p>Students' work in Unit 3 with scatter plots focuses on interpreting the slope and y-intercept of linear models fit to data and using those models to solve real-world problems as well as determining and interpreting the strength and direction of the correlation in real-world contexts. Instruction in this unit also includes formally fitting a linear function to data using technology. While this is included in instruction, it is not included on this assessment as students will not have access to such technology on state assessments.</p> <p>Students will build on this work in Unit 4 when working with linear functions in a variety of real-world context. Data from this assessment can be used to determine potential supports students may need when interpreting linear functions in context. For students needing remediation support for these benchmarks, consider using resources from Grade 8, Unit 11 to scaffold.</p> |
| <p style="text-align: center;"><b>Calculator Information</b></p>   | <p>The assessment is calculator neutral.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p style="text-align: center;"><b>Total Recommended Points</b></p> | <p>The total recommended points in this guide is 99 points. Teachers can either determine where best to use the extra point or teachers may round the total value up to 100 points.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Question                                                                                                               | 1a       | Benchmark(s)                             | MA.912.DP.2.4                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| The slope represents the change in total weight of the bowl and liquid, in grams, for each tablespoon of liquid added. |          | Recommended Scoring<br>(10 Total Points) | Consider no partial credit.                                                                                             |
| Question                                                                                                               | 1b       | Benchmark(s)                             | MA.912.DP.2.4                                                                                                           |
| The y-intercept represents the total weight of the bowl, in grams, before any liquid is added.                         |          | Recommended Scoring<br>(10 Total Points) | Consider no partial credit.                                                                                             |
| Question                                                                                                               | 1c       | Benchmark(s)                             | MA.912.DP.2.4                                                                                                           |
| Weight (g)                                                                                                             | 1127.32  | Recommended Scoring<br>(8 Total Points)  | Consider no partial credit.                                                                                             |
| Question                                                                                                               | 1d       | Benchmark(s)                             | MA.912.DP.2.4                                                                                                           |
| Amount of Liquid (tablespoons)                                                                                         | about 23 | Recommended Scoring<br>(8 Total Points)  | Consider no partial credit. Students may approximate differently, but should receive credit as long as it's reasonable. |

| Question                                                                                                  | 2                          | Benchmark(s)                                          | MA.912.DP.2.4                                                                                               |                                                        |                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                            | <b>Recommended Scoring</b><br><b>(5 Total Points)</b> | Consider no partial credit. Any trend line that follows the trend of the data should be considered correct. |                                                        |                                                                                                                                                                     |
| Question                                                                                                  | 2b                         | Benchmark(s)                                          | MA.912.DP.2.4                                                                                               |                                                        |                                                                                                                                                                     |
| <table border="1"><tr><td><b>Equation</b></td><td><math>y = \frac{100}{3}x + 160</math></td></tr></table> |                            | <b>Equation</b>                                       | $y = \frac{100}{3}x + 160$                                                                                  | <b>Recommended Scoring</b><br><b>(10 Total Points)</b> | Consider where the student drew the trend line for question 2a. Consider partial credit for students who have either the slope or y-intercept correct but not both. |
| <b>Equation</b>                                                                                           | $y = \frac{100}{3}x + 160$ |                                                       |                                                                                                             |                                                        |                                                                                                                                                                     |

| Question                                                                                                                                                                                                                                                                                                                                                                                             | 2c | Benchmark(s)                                           | MA.912.DP.2.4                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| The slope is the change in total sales, in millions of dollars, for each year since 2004.                                                                                                                                                                                                                                                                                                            |    | <b>Recommended Scoring</b><br><b>(10 Total Points)</b> | Consider no partial credit.                                                                                       |
| Question                                                                                                                                                                                                                                                                                                                                                                                             | 2d | Benchmark(s)                                           | MA.912.DP.2.4                                                                                                     |
| The $y$ -intercept represents the total sales, in millions of dollars, for the year 2004.                                                                                                                                                                                                                                                                                                            |    | <b>Recommended Scoring</b><br><b>(10 Total Points)</b> | Consider no partial credit.                                                                                       |
| Question                                                                                                                                                                                                                                                                                                                                                                                             | 3a | Benchmark(s)                                           | MA.912.DP.2.6                                                                                                     |
| <p>The data for The Netherlands has a</p> <div> <input type="radio"/> positive         <input type="radio"/> negative       </div> <p>correlation because</p> <div> <input type="radio"/> the slope of the line of best fit is         <input type="radio"/> most of the residual values are       </div> <div> <input type="radio"/> positive.         <input type="radio"/> negative.       </div> |    | <b>Recommended Scoring</b><br><b>(10 Total Points)</b> | Students should complete all pieces of the statement correctly to show understanding. Consider no partial credit. |

| Question                                                                                                | 3b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Benchmark(s)                                                                                                              | MA.912.DP.2.6                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                         | <p> <input type="radio"/> Germany<br/> <input type="radio"/> The Netherlands         </p> <p>has a</p> <p> <input type="radio"/> weaker<br/> <input type="radio"/> stronger         </p> <p>correlation than</p> <p> <input type="radio"/> Germany<br/> <input type="radio"/> The Netherlands         </p> <p>because the residuals have a</p> <p> <input type="radio"/> larger<br/> <input type="radio"/> smaller         </p> <p>absolute value, indicating that the</p> <p>distance each</p> <p> <input type="radio"/> actual<br/> <input type="radio"/> predicted         </p> <p>value is from the line of best fit is</p> <p> <input type="radio"/> larger.<br/> <input type="radio"/> smaller.         </p> <p style="text-align: center; color: red;">OR</p> <p> <input type="radio"/> Germany<br/> <input type="radio"/> The Netherlands         </p> <p>has a</p> <p> <input type="radio"/> weaker<br/> <input type="radio"/> stronger         </p> <p>correlation than</p> <p> <input type="radio"/> Germany<br/> <input type="radio"/> The Netherlands         </p> <p>because the residuals have a</p> <p> <input type="radio"/> larger<br/> <input type="radio"/> smaller         </p> <p>absolute value, indicating that the</p> <p>distance each</p> <p> <input type="radio"/> actual<br/> <input type="radio"/> predicted         </p> <p>value is from the line of best fit is</p> <p> <input type="radio"/> larger.<br/> <input type="radio"/> smaller.         </p> | <p style="text-align: center;"><b>Recommended Scoring</b></p> <p style="text-align: center;"><b>(10 Total Points)</b></p> | <p>There are two possible correct answers. Students should complete all pieces of the statement correctly to show understanding. Consider no partial credit.</p> |
| Question                                                                                                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Benchmark(s)                                                                                                              | MA.912.DP.2.4                                                                                                                                                    |
| <div style="background-color: #cccccc; padding: 5px; display: inline-block;"><b>Price in 2018</b></div> | <div style="border: 1px solid black; padding: 5px; display: inline-block;">\$10.81</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p style="text-align: center;"><b>Recommended Scoring</b></p> <p style="text-align: center;"><b>(8 Total Points)</b></p>  | <p>Consider no partial credit.</p>                                                                                                                               |